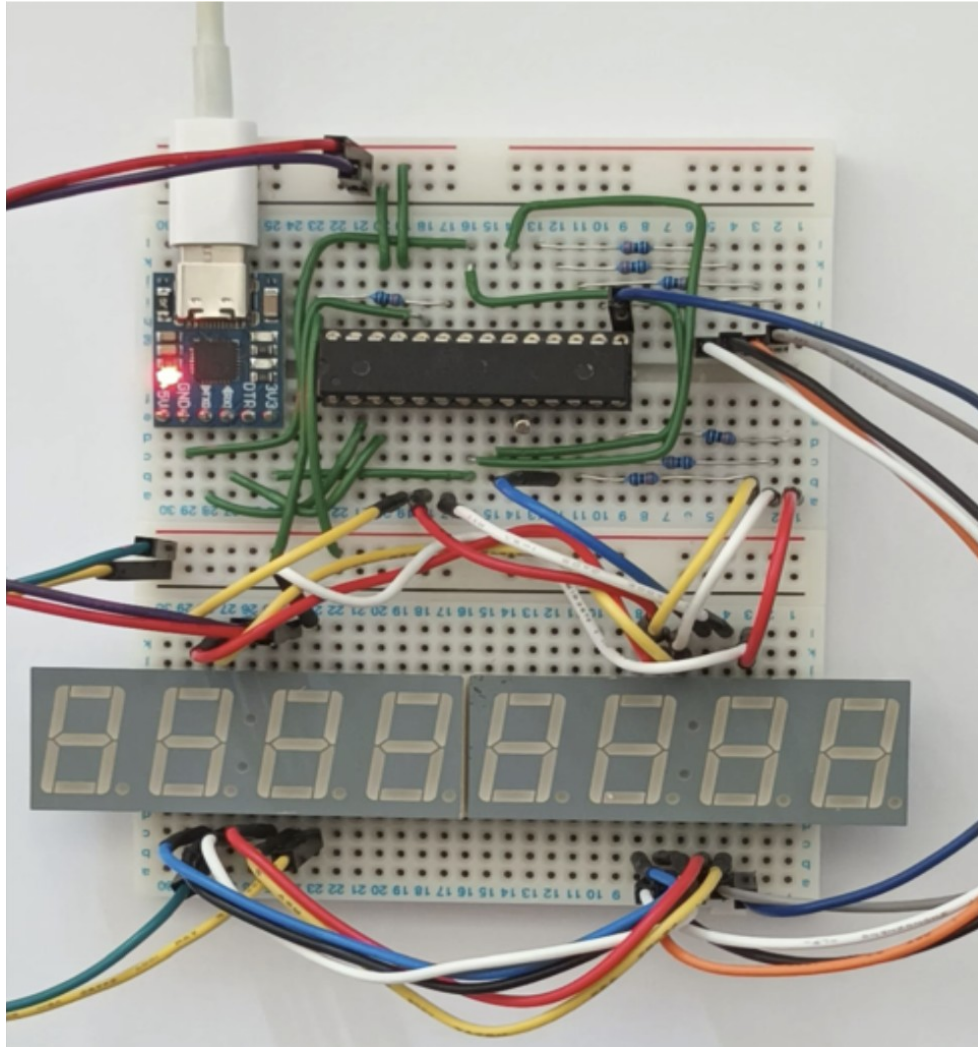


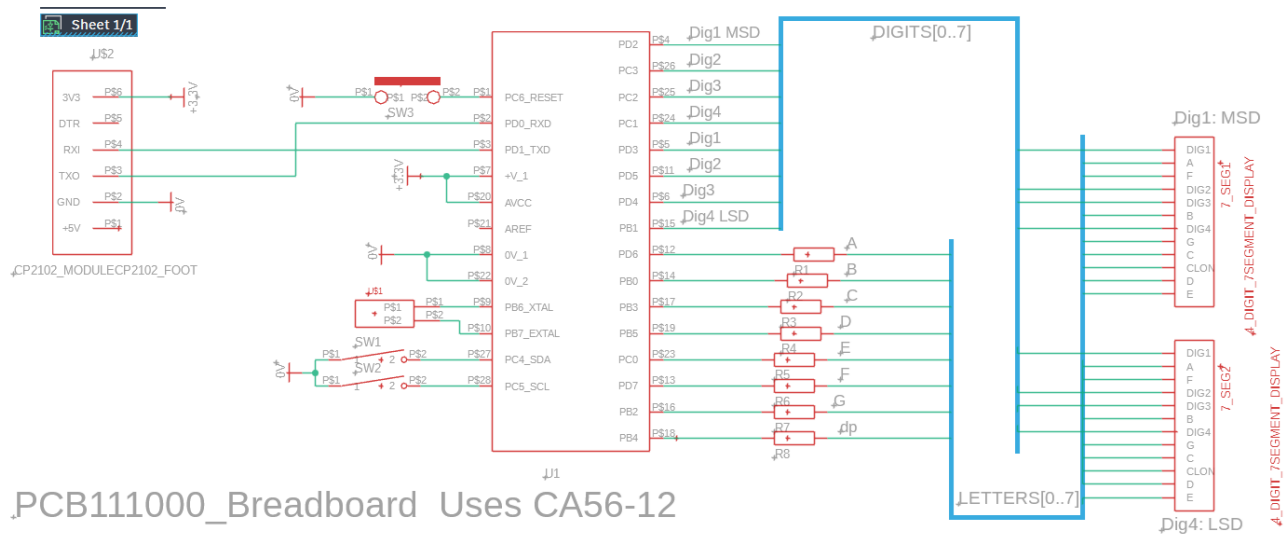
PCB111000_Breadboard and Breadboard_V2 Build

DIY version of PCB111000

PCB111000 uses an Atmega 328 and V2 uses an Atmega 168



Wiring Diagram



Extra components

Place 0.1uF caps between the switched inputs PC4 & 5 and 0V

Place a 1uF cap between pins 20 and 22 (AVCC and 0V)

Port	IC	Pin number	Pin	Display	Pin Number
B0		14	B		7
B1		15	6_Dig_4 RH		
B2		16	G		5
B3		17	C		4
B4		18	DP		3
B5		19	D		2
B6		9	Xtal		
B7		10	Xtal		
C0		23	E		1
C1		24	6_Dig 4 LH		
25_C2		25	8_Dig 3 LH		
26_C3		26	9_Dig 2 LH		
C4		27	I2C		
C5		28	I2C		
C6		1	Reset		
D0		2	Rx		
D1		3	Tx		
4_D2		4	12_Dig 1 LH		
5_D3		5	12_Dig 1 RH		
6_D4		6	8_Dig 3 RH		
11_D5		11	9_Dig 2 RH		
D6		12	A		11
D7		13	F		10

The short display uses the RH display module

Components

Item	Num	Cost per assembly	Part numbers	
Breadboard	2	£12.9	RS 189-2277 OK (Sunhayato SAD-101) <i>Gives good contact</i> <i>Do not use cheaper versions</i>	One piece for two displays One piece for the IC, CP2102 module and resistors
Display	2	£12.8	RS 872-1954 CA56-12 OK	Two required Also available from Mouser
IC	1	£2.425	127-8258 Atmega 168	
IC socket	1	£1.37	674-2498 OK	
330R	8	£0.112	199-7758 OK	
Crystal	1	£0.218	RS 197-708	
2 pin header	1	£0.136	251-8086 OK	For use with crystal
6 pin header	1	£0.3	828-1591	For use with CP2102 module
CP2102 module	1	£2.00	From EBAY	Type-C USB to UART TTL Module 6Pin Serial Converter CP2102 STC
Jumper wires	1 bag	£6.51	505-3403 50mm	50mm x 10
Jumper wires	1 bag	£2.74	791-6463 150mm	150mm x 10
Jumper wires	1 bag	£1.37	555-922 150mm	150mm x 10
Jumper wires	1 bag	£7.78	505-3374 100m	100mm x 10

Note: The Atmega 168 can be replaced by an Atmega 328. However with the Atmega 328 the clock stability was poor. This was probably due to interference caused by the high speed switching edges and the use of long wires.

Alternative option: Replaced the Atmega device and CP2102 USB bridge with an Arduino Nano board.